Principal Component Analysis

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Key Concepts

1) Principal component analysis (PCA) finds the best $p < N$ set of bases for a random signal with N values. Here "best" means the bases that minimize the mean-squared error between the signal and its representation using p bases.

2) Principal component bases are the eigenvectors corresponding to the p largest eigenvectors of the data covariance matrix.

3) PCA properties:

a) The mean squared error is the sum of the $N - p$ smallest eigenvalues of the data covariance matrix.

b) All components (basis weight times the basis) are uncorrelated with one another.

c) The basis weights are uncorrelated with each other and the variance of the weights are equal to the corresponding eigenvalue.

4) If a signal with N values can be represented by $p < N$ bases and the signal is measured in white noise, then representing the signal using PCA with p bases reduces the noise level by a factor of N/p.

Video Screenshots

$(N \times 1)$ $\underline{x} = \sum_k a_k \underline{\Psi}_k$ bases $\underline{\Psi}_k$

PCA - find best (mean-squared error) $p < N$ bases for "family" of random data $\underline{x}$

$$\underline{x} \approx \sum_{k=1}^{p} a_k \underline{\Psi}_k = \begin{bmatrix} \underline{\Psi}_1 & \underline{\Psi}_2 & \cdots & \underline{\Psi}_p \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = \underline{\underline{\Psi}}_p \underline{a}_p$$

Squared error

$$e^2(\underline{\underline{\Psi}}_p | \underline{x}) = \min_{\underline{a}_p} |x - \underline{\underline{\Psi}}_p \underline{a}_p|^2$$

$\underline{\Psi}_k$ orthogonal unit norm

PCA problem

$$\min_{\underline{\underline{\Psi}}_p} E_{\underline{x}} \{ e^2(\underline{\underline{\Psi}}_p | \underline{x}) \} \quad \text{subj. to} \quad \underline{\underline{\Psi}}_p^H \underline{\underline{\Psi}}_p = \underline{\underline{I}}$$



$$\underline{\Psi}_p^H \underline{\Psi}_p = \underline{I} \Rightarrow \underline{\Psi}_\ell^H \underline{\Psi}_k = \begin{cases} 0 & \ell \neq k \\ 1 & \ell = k \end{cases}$$

Hence, $\underline{a}_p = \underline{\Psi}_p^H \underline{x}$

$$\begin{aligned} e^2(\underline{\Psi}_p | \underline{x}) &= |\underline{x} - \underline{\Psi}_p \underline{\Psi}_p^H \underline{x}|^2 \\ &= \underline{x}^H (\underline{I} - \underline{\Psi}_p \underline{\Psi}_p^H)^2 \underline{x} \\ &= \underline{x}^H (\underline{I} - \underline{\Psi}_p \underline{\Psi}_p^H) \underline{x} = \underline{x}^H \underline{x} \\ &\quad - (\underline{x}^H \underline{\Psi}_p)(\underline{\Psi}_p^H \underline{x}) \end{aligned}$$

PCA problem

$$\max_{\underline{\Psi}_k, k=1,\dots p} E\left\{ \sum_{k=1}^{p} (\underline{x}^H \underline{\Psi}_k)(\underline{\Psi}_k^H \underline{x}) \right\} = \max_{\underline{\Psi}_k, k=1,\dots p} \sum_{k=1}^{p} \underline{\Psi}_k^H \underline{R} \underline{\Psi}_k$$

subj to $\underline{\Psi}_\ell^H \underline{\Psi}_k = \begin{cases} 0 & \ell \neq k \\ 1 & \ell = k \end{cases}$ $\qquad \underline{R} = E\{\underline{x}\underline{x}^H\}$

Solution: set $\underline{\Psi}_k$ to the eigenvector of $\underline{R}$ corresponding to the kth largest eigenvalue $\qquad \underline{R}\underline{\Psi}_k = \lambda_k \underline{\Psi}_k$

Properties –

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1) Mean Squared Error: $E\{e^2(\underline{\Psi}_p|\underline{x})\} = \sum_{k=p+1}^{N} \lambda_k$

"best" bases for explaining variance of $\underline{x}$

2) Uncorrelated Components $\underline{\Psi}_i a_i$ $\quad \underline{x} = \sum_{i=1}^{p} (\underline{\Psi}_i a_i)$

$$E\{a_i^* \underline{\Psi}_i^H \underline{\Psi}_k a_k\} = \begin{cases} 0 & i \neq k \\ E\{|a_k|^2\} & i = k \end{cases}$$

3) Uncorrelated Coefficients

$$E\{a_i^* a_k\} = \underline{\Psi}_i^H E\{\underline{x}\,\underline{x}^H\}\underline{\Psi}_k = \underline{\Psi}_i^H \underline{R}\,\underline{\Psi}_k = \underline{\Psi}_i^H \underline{\Psi}_k \lambda_k$$

$$= \begin{cases} 0 & i \neq k \\ \lambda_k & i = k \end{cases}$$

Low-rank Signals in White Noise

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Let $\underline{x} = \underline{s} + \underline{n}$ where $E\{\underline{n}\underline{n}^T\} = \sigma^2 \underline{I}$, $\underline{s} = \sum_{i=1}^{p} \underline{\psi}_i \alpha_i$

$= \underline{\Psi}_p \underline{\alpha}$

Consider $\hat{\underline{s}} = \underline{\Psi}_p \underline{a}_p = \underline{\Psi}_p \underline{\Psi}_p^T \underline{x}$

$$E\{|\hat{\underline{s}} - \underline{s}|^2\} = E\{|\underline{\Psi}_p \underline{\Psi}_p^T \underline{s} - \underline{s} + \underline{\Psi}_p \underline{\Psi}_p^T \underline{n}|^2\}$$

$$= E\{|\underline{\Psi}_p \underline{\Psi}_p^T \underline{n}|^2\} = E\{|\underline{n}^T \underline{\Psi}_p \underline{\Psi}_p^T \underline{\Psi}_p \underline{\Psi}_p^T \underline{n}|^2\}$$

$$= \sum_{i=1}^{p} E\{|\underline{\psi}_i^T \underline{n}|^2\} = \sum_{i=1}^{p} \sigma^2 = p\sigma^2$$

Compare to $E\{|x - s|^2\} = E\{|\underline{n}|^2\} = N\sigma^2$

PCA reduces white noise by factor $\frac{N}{p}$

Note: $\underline{R}_x = \underline{R}_s + \sigma^2 \underline{I} \Rightarrow \lambda_{xi} = \lambda_{si} + \sigma^2$

Example:

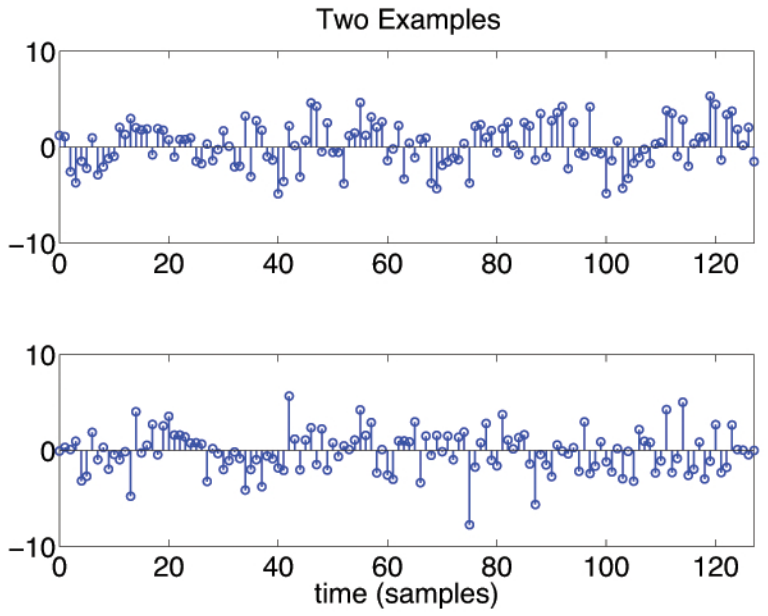
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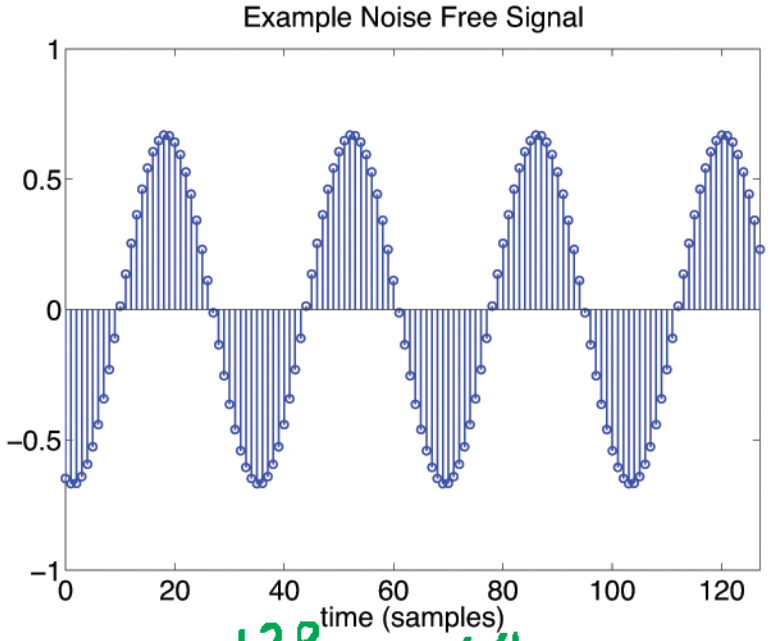
Random Amplitude/Phase Sinusoids in White Noise

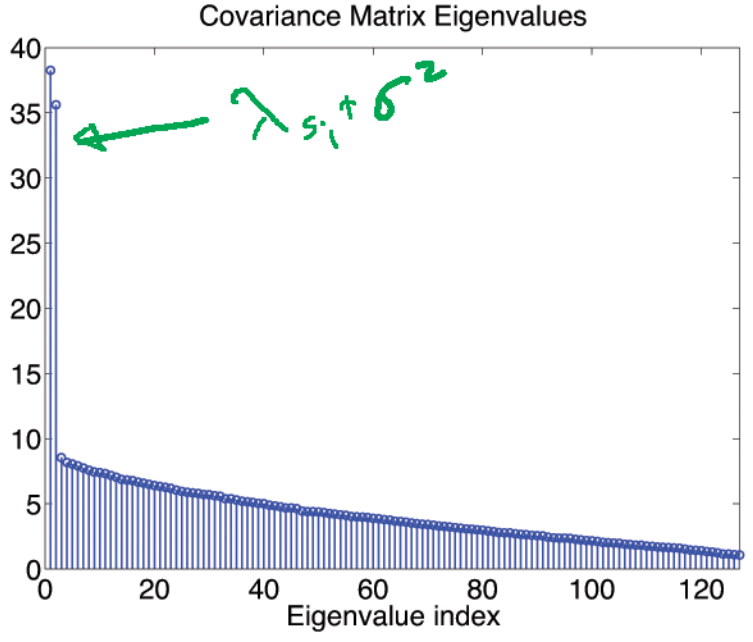
$\sigma^2 = 4$, $Amp \sim N(0,1)$

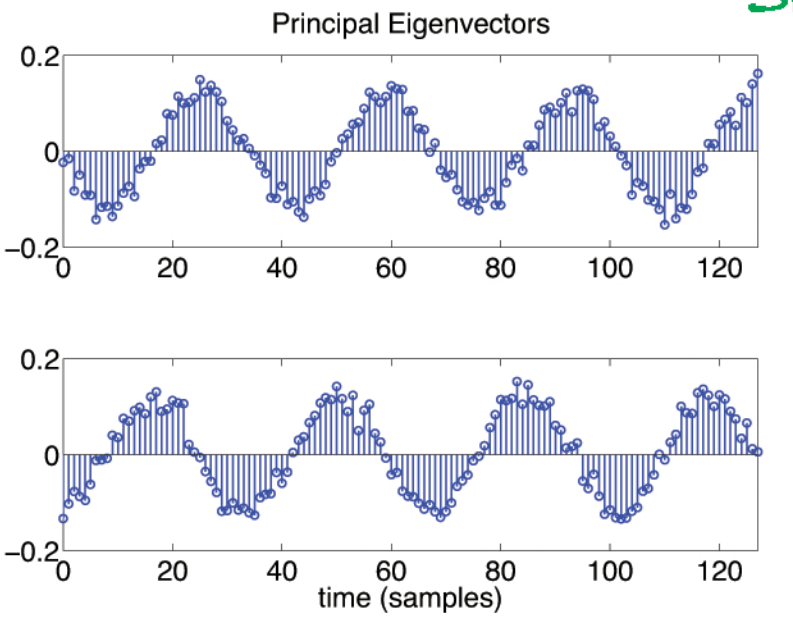
$Phase \sim U(0, 2\pi)$

$$R = \frac{1}{500} \sum_{i=1}^{500} \underline{x}_i \underline{x}_i^T \qquad N = 128$$

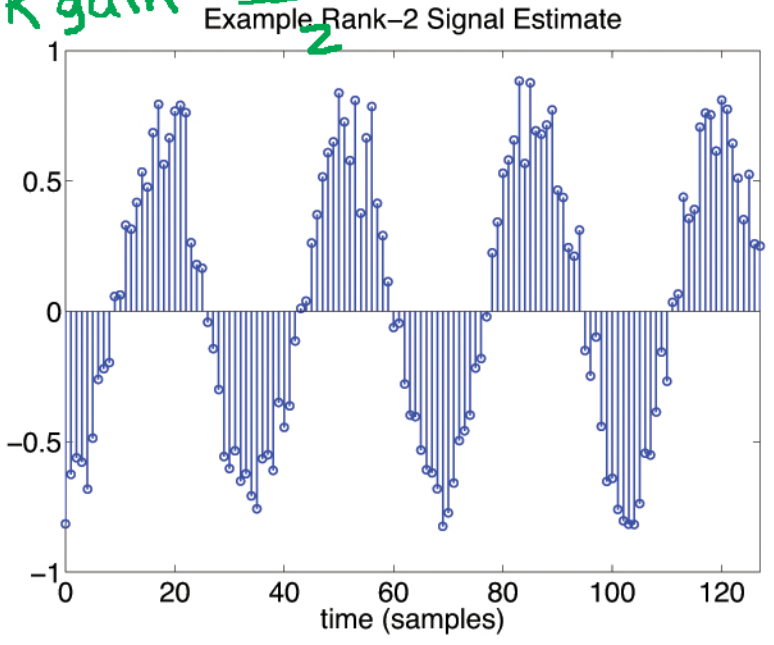








$$SNR\ gain = \frac{128}{2} = 64$$



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